Ho Chi Minh City University of Technology




Faculty of Computer Science and Engineering
Course: Computer Architecture Lab (CO2008)

# Lab 2

## Branches and procedures

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# 1 Introduction

- **The main purpose of this session is to get familiar with conditional branch, unconditional jump instructions, and call procedures.**
- Students are also expected to be able to work with recursion in this lab.
- Students must submit their answers to the LMS system no later than the last period of the lab section. Then, the instructor will evaluate all students' work during the lab section's final period.

# 2 Exercises

## 2.1 Exercise 1

Write a MIPS program with the following requirements:

1. Declare an integer array of width 5 with data entered from the keyboard.
2. Check whether each element in the array is divisible by 4 or not.
3. If an element is not divisible by 4, change it to the number divisible by 4 that is closest to it (round down is prioritized). For example, if the number is 35, the result will be 36, if the number is 34 or 33, the result will be 32.
4. Print the result to the terminal.

## 2.2 Exercise 2

Write a MIPS program that requests two positive integers called a and b from the user (please check if they are positive or not). The program then prints their greatest common divisor (GCD) and lowest common multiple (LCM) utilizing recursive. The input numbers should be after the prompt: "a = " and "b = ". For example, if the user input **a=6** and **b=45**, the result should be **GCD=3, LCM=90**.

## 2.3 Exercise 3

Write a MIPS program to find the second largest element in a 10-elements integer array. The program is also required to find all their indexes. Print the value and all of its indexes. The elements of the array must be inserted by the user after the prompt: "Please insert a element:"

For example, if the array is **1, 2, 7, -7, 3, 7, 4, 6, 8, 7** the output should be **Second largest value is 7, found in index 2, 5, 9**.

## 2.4 Exercise 4

Write a MIPS program to take a non-negative integer from the user (you need to check if the user input the correct number). Print the factorial number of that integer into the terminal.

For example, if the user input **5**, the output should be **120**. If the user input **-2**, the output should be **A factorial for your number can not be found.**